

**Golden Valley Educational Trust**  
**Dr. T. THIMMAIAH INSTITUTE OF TECHNOLOGY**

Approved by AICTE, Govt. of India, New Delhi  
Affiliated to Visvesvaraya Technological University, Belagavi.  
Accredited by NBA(CSE,ECE,EEE,ME & MNE Programs)  
Accredited by NAAC with 'A' Grade Oorgaum, Kolar Gold Fields-563120



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

VI Semester

**MACHINE LEARNING LABORATORY(BCSL606)**

**LAB MANUAL**

Name of the Student \_\_\_\_\_

USN Of the Student \_\_\_\_\_



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## **Department of Computer Science and Engineering**

### **Machine Learning Lab (BCSL606)**

#### **Lab Manual**

**Name of the Student:** \_\_\_\_\_

**USN:** \_\_\_\_\_

**Prepared by:**

1. Prof. Shalini G
2. Prof. Divya Shree V



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## Certificate

This is to certify that ..... Bearing  
USN: ..... has satisfactorily completed the experiments prescribed  
by the Visvesvaraya Technological University for the 6<sup>th</sup> semester B.E degree in the  
course **Machine Learning Lab** with course code **BCSL606** in the Laboratory of this  
college in the year 2025-26.

**Name & Signature of the Course Instructor**

**Head of the Department**

**Final Marks obtained**

50

## **Institution Vision and Mission**

### **Vision:**

To produce technically competent engineers having innovative skills, positive attitude, leadership and professional ethics, with a focus on developing sustainable and new technology

### **Mission:**

- Create a conducive environment for teaching, learning and innovation by providing state of the art infrastructure.
- Learn sustainable and new technologies through industry institute collaborations.
- Produce technically competent engineers with ethics, discipline and social consciousness through holistic education.

## **Department Vision and Mission**

### **Vision:**

To produce highly competent and innovative Computer Science professionals through excellence in teaching, training and research.

### **Mission:**

- M1: To provide appropriate infrastructure to impart need-based technical education through effective teaching and research.
- M2: To involve the students in innovative projects on emerging technologies to fulfill the industrial requirements.
- M3: To render leadership skills and ethical responsibilities in students that leads them to become globally competent professionals.

**Program Outcomes (POs)**

- PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
- PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
- PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
- PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for  
i) independent and life-long learning  
ii) adaptability to new and emerging technologies and  
iii) critical thinking in the broadest context of technological change. (WK8).

## **COURSE OBJECTIVES AND OUTCOMES**

**Course Learning Objectives:** This course will enable students to:

|             |                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLO1</b> | To become familiar with data and visualize univariate, bivariate, and multivariate data using statistical techniques and dimensionality reduction. |
| <b>CLO2</b> | To understand various machine learning algorithms such as similarity-based learning, regression, decision trees, and clustering.                   |
| <b>CLO3</b> | To familiarize with learning theories, probability-based models and developing the skills required for decision-making in dynamic environments.    |

**Course Outcomes:** At the end of the course the student will be able to:

|            |                                                                                                               |
|------------|---------------------------------------------------------------------------------------------------------------|
| <b>CO1</b> | Apply data visualization techniques to visualize relationship between the features and reduce the dimensions. |
| <b>CO2</b> | To Solve regression and classification tasks using ML algorithms.                                             |
| <b>CO3</b> | To implement clustering algorithms.                                                                           |

### **Assessment Details (CIE and SEE) (Course Specific)**

- The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%.
- The minimum passing marks for the CIE is 40% of the maximum marks (20 marks out of 50).
- The minimum passing marks for the SEE is 35% of the maximum marks (18marks out of 50).
- A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/course if the student secures not less than 35% (18 Marks out of 50) in the semester-end examination (SEE), and a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

### **Continuous Internal Evaluation (CIE):**

- CIE marks for the practical course are 50 Marks.

## Rubrics for Lab Performance Assessment

The students will be assessed on their participation and performance in every lab. The marks obtained in each lab will be added and combined with Final Lab test or Viva-voce. The students need to attend 85% of the labs to qualify to be eligible to attend the university examination.

### Core Lab Participation Rubric (Conduction& Record)

| <b>Proficient<br/>(20-17)</b>                                                                                                                                                                                                                                                                                                                             | <b>Good<br/>(16-14)</b>                                                                                                                                                                                | <b>Adequate<br/>(13-10)</b>                                                                                                                                                                                                                              | <b>Substandard<br/>(9-6)</b>                                                                                                                                                                                                                 | <b>Unacceptable<br/>(5-0)</b>                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Lab performance is excellent.</p> <p>The student has attended all labs.</p> <p>Student demonstrates an accurate understanding of the lab objectives and concepts.</p> <p>The student can correctly answer questions and if appropriate, can explain concepts to fellow classmates.</p> <p>Student is eager to participate and assists when needed.</p> | <p>Lab performance is good. At most, the student has one lab absence.</p> <p>Student demonstrates a fair understanding of the lab objectives and concepts.</p> <p>The student can answer questions</p> | <p>Lab performance is adequate.</p> <p>The student is absent more than 2 classes.</p> <p>Student arrives on time to lab, but may be unprepared.</p> <p>Answers to questions are basic and superficial suggesting that concepts are not fully grasped</p> | <p>Lab performance is fair.</p> <p>The student may have been absent 3-4times.</p> <p>Student unpreparedness makes it impossible to fully participate.</p> <p>If able to participate, student has difficulty explaining key lab concepts.</p> | <p>Lab performance is barely adequate.</p> <p>The student is irregular.</p> <p>Student was late to lab or did not participate.</p> <p>There was no attempt to make prior arrangements to make up the lab.</p> |

| Machine Learning lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Semester   | 6   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | BCSL606                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CIE Marks  | 50  |
| Teaching Hours/Week (L:T:P: S)                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0:0:2:0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SEE Marks  | 50  |
| Credits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Exam Hours | 100 |
| Examination type (SEE)                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Practical                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |     |
| <b>Course objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |     |
| <ul style="list-style-type: none"><li>To become familiar with data and visualize univariate, bivariate, and multivariate data using statistical techniques and dimensionality reduction.</li><li>To understand various machine learning algorithms such as similarity-based learning, regression, decision trees, and clustering.</li><li>To familiarize with learning theories, probability-based models and developing the skills required for decision-making in dynamic environments.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |     |
| <b>Sl.NO</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Experiments</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |     |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Develop a program to create histograms for all numerical features and analyze the distribution of each feature. Generate box plots for all numerical features and identify any outliers. Use California Housing dataset.<br><br><b>Book 1: Chapter 2</b>                                                                                                                                                                                                                                                   |            |     |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Develop a program to Compute the correlation matrix to understand the relationships between pairs of features. Visualize the correlation matrix using a heatmap to know which variables have strong positive/negative correlations. Create a pair plot to visualize pairwise relationships between features. Use California Housing dataset.<br><br><b>Book 1: Chapter 2</b>                                                                                                                               |            |     |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Develop a program to implement Principal Component Analysis (PCA) for reducing the dimensionality of the Iris dataset from 4 features to 2.<br><br><b>Book 1: Chapter 2</b>                                                                                                                                                                                                                                                                                                                                |            |     |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | For a given set of training data examples stored in a .CSV file, implement and demonstrate the Find-S algorithm to output a description of the set of all hypotheses consistent with the training examples.<br><br><b>Book 1: Chapter 3</b>                                                                                                                                                                                                                                                                |            |     |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Develop a program to implement k-Nearest Neighbour algorithm to classify the randomly generated 100 values of $x$ in the range of $[0,1]$ . Perform the following based on dataset generated.<br><br>a. Label the first 50 points $\{x_1, \dots, x_{50}\}$ as follows: if $(x_i \leq 0.5)$ , then $x_i \in \text{Class}_1$ , else $x_i \in \text{Class}_2$<br>b. Classify the remaining points, $x_{51}, \dots, x_{100}$ using KNN. Perform this for $k=1,2,3,4,5,20,30$<br><br><b>Book 2: Chapter – 2</b> |            |     |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs<br><br><b>Book 1: Chapter – 4</b>                                                                                                                                                                                                                                                                                                          |            |     |
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Develop a program to demonstrate the working of Linear Regression and Polynomial Regression. Use Boston Housing Dataset for Linear Regression and Auto MPG Dataset (for vehicle fuel efficiency prediction) for Polynomial Regression.<br><br><b>Book 1: Chapter – 5</b>                                                                                                                                                                                                                                   |            |     |
| 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Develop a program to demonstrate the working of the decision tree algorithm. Use Breast Cancer Data set for building the decision tree and apply this knowledge to classify a new sample.<br><br><b>Book 2: Chapter – 3</b>                                                                                                                                                                                                                                                                                |            |     |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Develop a program to implement the Naive Bayesian classifier considering Olivetti Face Data set for training. Compute the accuracy of the classifier, considering a few test data sets.<br><b>Book 2: Chapter – 4</b> |
| 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Develop a program to implement k-means clustering using Wisconsin Breast Cancer data set and visualize the clustering result.<br><b>Book 2: Chapter – 4</b>                                                           |
| <b>Course outcomes (Course Skill Set):</b><br>At the end of the course the student will be able to: <ul style="list-style-type: none"> <li>• Illustrate the principles of multivariate data and apply dimensionality reduction techniques.</li> <li>• Demonstrate similarity-based learning methods and perform regression analysis.</li> <li>• Develop decision trees for classification and regression problems, and Bayesian models for probabilistic learning.</li> <li>• Implement the clustering algorithms to share computing resources.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                       |
| <b>Assessment Details (both CIE and SEE)</b><br><p>The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together</p> <p><b>Continuous Internal Evaluation (CIE):</b><br/>         CIE marks for the practical course are <b>50 Marks</b>.<br/>         The split-up of CIE marks for record/ journal and test are in the ratio <b>60:40</b>.</p> <ul style="list-style-type: none"> <li>• Each experiment is to be evaluated for conduction with an observation sheet and record write-up. Rubrics for the evaluation of the journal/write-up for hardware/software experiments are designed by the faculty who is handling the laboratory session and are made known to students at the beginning of the practical session.</li> <li>• Record should contain all the specified experiments in the syllabus and each experiment write-up will be evaluated for 10 marks.</li> <li>• Total marks scored by the students are scaled down to <b>30 marks</b> (60% of maximum marks).</li> <li>• Weightage to be given for neatness and submission of record/write-up on time.</li> <li>• Department shall conduct a test of 100 marks after the completion of all the experiments listed in the syllabus.</li> <li>• In a test, test write-up, conduction of experiment, acceptable result, and procedural knowledge will carry a weightage of 60% and the rest 40% for viva-voce.</li> <li>• The suitable rubrics can be designed to evaluate each student's performance and learning ability.</li> <li>• The marks scored shall be scaled down to <b>20 marks</b> (40% of the maximum marks).</li> </ul> <p>The Sum of scaled-down marks scored in the report write-up/journal and marks of a test is the total CIE marks scored by the student.</p> |                                                                                                                                                                                                                       |

**Semester End Evaluation (SEE):**

- SEE marks for the practical course are 50 Marks.
- SEE shall be conducted jointly by the two examiners of the same institute, examiners are appointed by the Head of the Institute.
- The examination schedule and names of examiners are informed to the university before the conduction of the examination. These practical examinations are to be conducted between the schedule mentioned in the academic calendar of the University.
- All laboratory experiments are to be included for practical examination.
- (Rubrics) Breakup of marks and the instructions printed on the cover page of the answer script to be strictly adhered to by the examiners. **OR** based on the course requirement evaluation rubrics shall be decided jointly by examiners.
- Students can pick one question (experiment) from the questions lot prepared by the examiners jointly.
- Evaluation of test write-up/ conduction procedure and result/viva will be conducted jointly by examiners.

General rubrics suggested for SEE are mentioned here, writeup-20%, Conduction procedure and result in -60%, Viva-voce 20% of maximum marks. SEE for practical shall be evaluated for 100 marks and scored marks shall be scaled down to 50 marks (however, based on course type, rubrics shall be decided by the examiners)

Change of experiment is allowed only once and 15% of Marks allotted to the procedure part are to be made zero.

The minimum duration of SEE is 02 hours

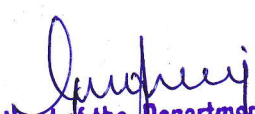
**Suggested Learning Resources:**

**Books:**

1. S Sridhar and M Vijayalakshmi, "Machine Learning", Oxford University Press, 2021.
2. M N Murty and Ananthanarayana V S, "Machine Learning: Theory and Practice", Universities Press (India) Pvt. Limited, 2024.

**Web links and Video Lectures (e-Resources):**

- [https://www.drssridhar.com/?page\\_id=1053](https://www.drssridhar.com/?page_id=1053)
- <https://www.universitiespress.com/resources?id=9789393330697>
- [https://onlinecourses.nptel.ac.in/noc23\\_cs18/preview](https://onlinecourses.nptel.ac.in/noc23_cs18/preview)

  
**Head of the Department**  
Department of Computer Science and Engineering  
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Dorgaam, Kolar Gold Field, Karnataka - 563120

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**Department of Computer Science and Engineering Machine**  
**Machine Learning Lab**  
**(BCSL606)**

**List of Experiments**

| <b>Exp. No</b> | <b>Title of the Experiment</b>                                                                                                                                                                                                                                                                                                                                                                              | <b>Page No</b> |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1.             | Develop a program to create histograms for all numerical features and analyze the distribution of each feature. Generate box plots for all numerical features and identify any outliers. Use California Housing dataset.                                                                                                                                                                                    | 09             |
| 2.             | Develop a program to Compute the correlation matrix to understand the relationships between pairs of features. Visualize the correlation matrix using a heatmap to know which variables have strong positive/negative correlations. Create a pair plot to visualize pairwise relationships between features. Use California Housing dataset.                                                                | 12             |
| 3.             | Develop a program to implement Principal Component Analysis (PCA) for reducing the dimensionality of the Iris dataset from 4 features to 2.                                                                                                                                                                                                                                                                 | 15             |
| 4.             | For a given set of training data examples stored in a .CSV file, implement and demonstrate the Find-S algorithm to output a description of the set of all hypotheses consistent with the training examples.                                                                                                                                                                                                 | 18             |
| 5.             | Develop a program to implement k-Nearest Neighbour algorithm to classify the randomly generated 100 values of x in the range of [0,1]. Perform the following based on dataset generated. a. Label the first 50 points {x1,.....,x50} as follows: if ( $x_i \leq 0.5$ ), then $x_i$ Class1, else $x_i$ Class1 b. Classify the remaining points, x51,.....,x100 using KNN. Perform this for k=1,2,3,4,5,20,30 | 21             |
| 6.             | Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs                                                                                                                                                                                                                                             | 25             |
| 7.             | Develop a program to demonstrate the working of Linear Regression and Polynomial Regression. Use Boston Housing Dataset for Linear Regression and Auto MPG Dataset (for vehicle fuel efficiency prediction) for Polynomial Regression.                                                                                                                                                                      | 28             |

|     |                                                                                                                                                                                           |    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 8.  | Develop a program to demonstrate the working of the decision tree algorithm. Use Breast Cancer Data set for building the decision tree and apply this knowledge to classify a new sample. | 32 |
| 9.  | Develop a program to implement the Naive Bayesian classifier considering Olivetti Face Data set for training. Compute the accuracy of the classifier, considering a few test data sets.   | 36 |
| 10. | Develop a program to implement k-means clustering using Wisconsin Breast Cancer data set and visualize the clustering result.                                                             | 39 |

## Evaluation Sheet (for Core Course)

| Sl. No | Title of the Experiment | Conduction Date | Submission Date | Conduction (20) | Record (10) | Course Instructor Sign |
|--------|-------------------------|-----------------|-----------------|-----------------|-------------|------------------------|
| 1.     |                         |                 |                 |                 |             |                        |
| 2.     |                         |                 |                 |                 |             |                        |
| 3.     |                         |                 |                 |                 |             |                        |
| 4.     |                         |                 |                 |                 |             |                        |
| 5.     |                         |                 |                 |                 |             |                        |
| 6.     |                         |                 |                 |                 |             |                        |
| 7.     |                         |                 |                 |                 |             |                        |
| 8.     |                         |                 |                 |                 |             |                        |
| 9.     |                         |                 |                 |                 |             |                        |
| 10.    |                         |                 |                 |                 |             |                        |
| 11.    |                         |                 |                 |                 |             |                        |
|        |                         |                 |                 | Average         |             |                        |

| Avg Conduction Marks (20) | Avg Record Marks (10) | Total Average (30) | Test Marks (20) | Total (50) | Faculty Signature with Date |
|---------------------------|-----------------------|--------------------|-----------------|------------|-----------------------------|
|                           |                       |                    |                 |            |                             |

**Experiment 1.****Date:**

**AIM: Develop a program to create histograms for all numerical features and analyze the distribution of each feature. Generate box plots for all numerical features and identify any outliers. Use California Housing dataset.**

**Solution:****Import Necessary Libraries**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
df = pd.read_csv(r"C:\Users\Asus\Documents\Datasets[1]\Datasets\housing.csv")
df.head()
df.shape
df.info()
df.nunique()
```

**Data Cleaning**

```
df.isnull().sum()
df.duplicated().sum()
df['total_bedrooms'].median()
# Handling missing values
df['total_bedrooms'].fillna(df['total_bedrooms'].median(), inplace=True)
```

**Feature Engineering**

```
for i in df.iloc[:,2:7]:
```

```
df[i] = df[i].astype('int')
```

### **Descriptive Statistics**

```
df.describe().T
```

```
Numerical = df.select_dtypes(include=[np.number]).columns
```

```
print(Numerical)
```

### **Uni-Variate Analysis**

```
for col in Numerical:
```

```
    plt.figure(figsize=(10, 6))
```

```
    df[col].plot(kind='hist', title=col, bins=60, edgecolor='black')
```

```
    plt.ylabel('Frequency')
```

```
    plt.show()
```

```
for col in Numerical:
```

```
    plt.figure(figsize=(6, 6))
```

```
sns.boxplot(df[col], color='blue')
```

```
plt.title(col)
```

```
plt.ylabel(col)
```

```
plt.show()
```

**VIVA Questions with Answers after each experiment****1. What is the California Housing dataset?**

It is a standard dataset containing housing-related numerical features of California districts, commonly used for data analysis and machine learning tasks.

**2. Why are histograms used in data analysis?**

Histograms are used to understand the distribution of numerical features such as whether the data is normal, skewed, or uniform.

**3. What information does a box plot provide?**

A box plot shows the median, quartiles, data spread, and helps identify outliers in a dataset.

**4. What are numerical features in the California Housing dataset?**

Numerical features include median income, house age, total rooms, total bedrooms, population, households, and median house value.

**5. How do histograms help in feature analysis?**

They help visualize frequency distribution and reveal skewness, peaks, and data concentration.

**6. What is an outlier?**

An outlier is a data point that significantly differs from the rest of the data values.

**7. How are outliers identified using box plots?**

Outliers appear as points lying outside the whiskers of the box plot.

**8. Why is outlier detection important?**

Outliers can affect statistical analysis and machine learning model performance if not handled properly.

**9. Which Python libraries are commonly used to create histograms and box plots?**

Matplotlib and Seaborn are commonly used for creating histograms and box plots.

**10. What observation can be made from the California Housing dataset distributions?**

Many features show right-skewed distributions and contain outliers, especially population and total rooms.

**Experiment 2.****Date:**

**AIM: Develop a program to Compute the correlation matrix to understand the relationships between pairs of features. Visualize the correlation matrix using a heatmap to know which variables have strong positive/negative correlations. Create a pair plot to visualize pairwise relationships between features. Use California Housing dataset.**

**Solution:**

```
import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
```

**# Step 1: Load the California Housing Dataset**

```
california_data = fetch_california_housing(as_frame=True)

data = california_data.frame
```

**# Step 2: Compute the correlation matrix**

```
correlation_matrix = data.corr()
```

**# Step 3: Visualize the correlation matrix using a heatmap**

```
plt.figure(figsize=(10, 8))

sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt='.2f', linewidths=0.5)

plt.title('Correlation Matrix of California Housing Features')

plt.show()
```

**# Step 4: Create a pair plot to visualize pairwise relationships**

```
sns.pairplot(data, diag_kind='kde', plot_kws={'alpha': 0.5})  
  
plt.suptitle('Pair Plot of California Housing Features', y=1.02)  
  
plt.show()
```

**VIVA Questions with Answers after each experiment****1. What is a correlation matrix?**

A correlation matrix is a table that shows correlation coefficients between pairs of numerical features.

**2. What does correlation measure?**

Correlation measures the strength and direction of the relationship between two variables.

**3. What is the range of correlation coefficient values?**

Correlation coefficient values range from **-1 to +1**.

**4. What does a positive correlation indicate?**

A positive correlation indicates that as one variable increases, the other variable also increases.

**5. What does a negative correlation indicate?**

A negative correlation indicates that as one variable increases, the other variable decreases.

**6. Why is a heatmap used to visualize the correlation matrix?**

A heatmap uses colors to easily identify strong and weak correlations between features.

**7. What information does a pair plot provide?**

A pair plot visualizes pairwise relationships and distributions of multiple features in one plot.

**8. Which features show strong correlation in the California Housing dataset?**

Median income shows a strong positive correlation with median house value.

**9. Which Python libraries are commonly used to create heatmaps and pair plots?**

Seaborn and Matplotlib are commonly used for heatmaps and pair plots.

**10. Why is correlation analysis important in data analysis?**

Correlation analysis helps in feature selection and understanding relationships between variables.

**Experiment 3.****Date:**

**AIM: Develop a program to implement Principal Component Analysis (PCA) for reducing the dimensionality of the Iris dataset from 4 features to 2.**

**Solution:**

```
# Import necessary libraries

from sklearn.datasets import load_iris

from sklearn.decomposition import PCA

from sklearn.preprocessing import StandardScaler

import pandas as pd

import matplotlib.pyplot as plt

# Step 1: Load the Iris dataset

iris = load_iris()

features = iris.data # The 4 features: Sepal Length, Sepal Width, Petal Length, Petal Width

target = iris.target # The target class (species)


# Step 2: Standardize the features

scaler = StandardScaler()

features_standardized = scaler.fit_transform(features)


# Step 3: Apply PCA to reduce to 2 components

pca = PCA(n_components=2)

features_pca = pca.fit_transform(features_standardized)


# Step 4: Create a DataFrame for the reduced data

pca_df = pd.DataFrame(data=features_pca, columns=["Principal Component 1", "Principal Component 2"])
```

```
pca_df["Target"] = target
```

### # Step 5: Visualize the results

```
plt.figure(figsize=(8, 6))
for label, color in zip(iris.target_names, ["red", "green", "blue"]):
    plt.scatter(
        pca_df.loc[pca_df["Target"] == list(iris.target_names).index(label), "Principal Component 1"],
        pca_df.loc[pca_df["Target"] == list(iris.target_names).index(label), "Principal Component 2"],
        label=label,
        alpha=0.7
    )
plt.title("PCA on Iris Dataset (4 features to 2 features)", fontsize=14)
plt.xlabel("Principal Component 1", fontsize=12)
plt.ylabel("Principal Component 2", fontsize=12)
plt.legend(title="Species")
plt.grid()
plt.show()
explained_variance = pca.explained_variance_ratio_
print("Explained Variance by each Principal Component:")
print("Principal Component 1: ",explained_variance[0])
print("Principal Component 2: ",explained_variance[1])
print("Total Variance Retained: ",sum(explained_variance))
```

**VIVA Questions with Answers after each experiment****1. What is Principal Component Analysis (PCA)?**

PCA is a dimensionality reduction technique that transforms correlated features into a smaller number of uncorrelated variables called principal components.

**2. Why is PCA used in data analysis?**

PCA is used to reduce dimensionality while preserving maximum variance in the data.

**3. How many features are present in the Iris dataset originally?**

The Iris dataset has 4 numerical features.

**4. To how many dimensions is the Iris dataset reduced in this program?**

The dataset is reduced from 4 features to 2 principal components.

**5. What are principal components?**

Principal components are new orthogonal axes that capture the maximum variance in the data.

**6. Why is data standardization important before applying PCA?**

Standardization ensures all features contribute equally to PCA by scaling them to the same range.

**7. Which principal component captures the maximum variance?**

The first principal component captures the maximum variance.

**8. Which Python library is commonly used to implement PCA?**

Scikit-learn is commonly used to implement PCA.

**9. What is the benefit of reducing features from 4 to 2?**

It simplifies visualization and reduces computational complexity.

**10. Can PCA be used for classification directly?**

No, PCA is mainly used for dimensionality reduction, not for classification.

**Experiment 4.****Date:**

**AIM: For a given set of training data examples stored in a .CSV file, implement and demonstrate the Find-S algorithm to output a description of the set of all hypotheses consistent with the training examples.**

**Solution:**

```
import pandas as pd

data = pd.read_csv(r"C:\Users\Asus\Desktop\training_data.csv")

print(data)

def find_s_algorithm(data):
    """Implements the Find-S algorithm to find the most specific hypothesis."""
    # Extract feature columns and target column
    attributes = data.iloc[:, :-1].values # All columns except last
    target = data.iloc[:, -1].values # Last column (class labels)

    # Step 1: Initialize hypothesis with first positive example
    for i in range(len(target)):
        if target[i] == "Yes": # Consider only positive examples
            hypothesis = attributes[i].copy()
            break

    # Step 2: Update hypothesis based on other positive examples
    for i in range(len(target)):
        if target[i] == "Yes":
            for j in range(len(hypothesis)):
                if hypothesis[j] != attributes[i][j]:
                    hypothesis[j] = '?' # Generalize inconsistent attributes
```

```
return hypothesis
```

**# Run Find-S Algorithm**

```
final_hypothesis = find_s_algorithm(data)
```

**# Print the learned hypothesis**

```
print("Most Specific Hypothesis:", final_hypothesis)
```

**VIVA Questions with Answers after each experiment****1. What is the Find-S algorithm?**

Find-S is a machine learning algorithm used to find the most specific hypothesis consistent with all positive training examples.

**2. Which type of learning does Find-S represent?**

Find-S represents **concept learning** in supervised machine learning.

**3. What type of training examples does Find-S use?**

Find-S uses only **positive training examples**.

**4. What is meant by a hypothesis in Find-S?**

A hypothesis is a conjunction of attribute constraints that describes the target concept.

**5. Why is a CSV file used in the Find-S algorithm?**

A CSV file is used to store and easily load structured training data.

**6. How does Find-S initialize the hypothesis?**

The hypothesis is initialized to the most specific hypothesis, usually represented by empty or null values.

**7. How is the hypothesis updated during learning?**

For each positive example, the hypothesis is generalized to match the example.

**8. What is the final output of the Find-S algorithm?**

The final output is the most specific hypothesis consistent with all positive examples.

**9. What is a limitation of the Find-S algorithm?**

Find-S ignores negative examples and may produce an incomplete hypothesis.

**10. Where is the Find-S algorithm mainly used?**

It is mainly used for teaching basic concept learning in machine learning.

```
hypothesis[j] = '?' # Generalize inconsistent attributes
```

```
return hypothesis
```

### **# Run Find-S Algorithm**

```
final_hypothesis = find_s_algorithm(data)
```

### **# Print the learned hypothesis**

```
print("Most Specific Hypothesis:", final_hypothesis)
```

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It is mainly used for teaching basic concept learning in machine learning.

**Experiment 5.****Date:**

**AIM: Develop a program to implement k-Nearest Neighbour algorithm to classify the randomly generated 100 values of x in the range of [0,1]. Perform the following based on dataset generated.**

**1. Label the first 50 points {x1,.....,x50} as follows: if ( $x_i \leq 0.5$ ), then  $x_i \in \text{Class1}$ , else  $x_i \in \text{Class1}$**

**2. Classify the remaining points, x51,.....,x100 using KNN. Perform this for k=1,2,3,4,5,20,30**

**Solution:**

```
import numpy as np

import matplotlib.pyplot as plt

from collections import Counter

data = np.random.rand(100)

labels = ["Class1" if x <= 0.5 else "Class2" for x in data[:50]]

def euclidean_distance(x1, x2):

    return abs(x1 - x2)

def knn_classifier(train_data, train_labels, test_point, k):

    distances = [(euclidean_distance(test_point, train_data[i]), train_labels[i]) for i in

range(len(train_data))]

    distances.sort(key=lambda x: x[0])

    k_nearest_neighbors = distances[:k]

    k_nearest_labels = [label for _, label in k_nearest_neighbors]

    return Counter(k_nearest_labels).most_common(1)[0][0]

train_data = data[:50]

train_labels = labels
```

```
test_data = data[50:]

k_values = [1, 2, 3, 4, 5, 20, 30]

print("--- k-Nearest Neighbors Classification ---")

print("Training dataset: First 50 points labeled based on the rule (x <= 0.5 -> Class1, x > 0.5 -> Class2)")

print("Testing dataset: Remaining 50 points to be classified\n")

results = {}

for k in k_values:

    print(f"Results for k = {k}:")

    classified_labels = [knn_classifier(train_data, train_labels, test_point, k) for test_point in test_data]

    results[k] = classified_labels

    for i, label in enumerate(classified_labels, start=51):

        print(f"Point x {i} (value: {test_data[i - 51]:.4f}) is classified as {label}")

    print("\n")

print("Classification complete.\n")

for k in k_values:

    classified_labels = results[k]

    class1_points = [test_data[i] for i in range(len(test_data)) if classified_labels[i] == "Class1"]

    class2_points = [test_data[i] for i in range(len(test_data)) if classified_labels[i] == "Class2"]

    plt.figure(figsize=(10, 6))

    plt.scatter(train_data, [0] * len(train_data), c=["blue" if label == "Class1" else "red" for label in train_labels],
```

```
label="Training Data", marker="o")  
  
plt.scatter(class1_points, [1] * len(class1_points), c="blue", label="Class1 (Test)",  
marker="x")  
  
plt.scatter(class2_points, [1] * len(class2_points), c="red", label="Class2 (Test)", marker="x")  
  
plt.title(f"k-NN Classification Results for k = {k}")  
  
plt.xlabel("Data Points")  
  
plt.ylabel("Classification Level")  
  
plt.legend()  
  
plt.grid(True)  
  
plt.show()
```

**VIVA Questions with Answers after each experiment****1. What is the k-Nearest Neighbour (k-NN) algorithm?**

k-NN is a supervised learning algorithm that classifies a data point based on the majority class of its k nearest neighbors.

**2. What type of learning algorithm is k-NN?**

k-NN is an instance-based and lazy learning algorithm.

**3. Why are random values generated in the range [0,1]?**

The range [0,1] simplifies distance calculation and helps demonstrate classification behavior clearly.

**4. How are the first 50 data points labeled?**

If  $x_i \leq 0.5$ , the point is labeled as **Class 1**; otherwise, it is labeled as **Class 2**.

**5. What is the purpose of using the first 50 points?**

They are used as training data for the k-NN classifier.

**6. How are the remaining points ( $x_{51}$  to  $x_n$ ) classified?**

They are classified using the k-NN algorithm based on the labeled training points.

**7. What distance measure is commonly used in this program?**

Euclidean distance is commonly used.

**8. How does the value of k affect classification?**

Small k values are sensitive to noise, while large k values produce smoother and more stable classification.

**9. Why are multiple k values (1, 2, 3, 4, 5, 20, 30) tested?**

To observe how different k values influence classification accuracy and decision boundaries.

**10. What is a limitation of the k-NN algorithm?**

k-NN has high computation cost during prediction and is sensitive to the choice of k.

**Experiment 6.****Date:**

**AIM: Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs.**

**Solution:**

```
import numpy as np

import matplotlib.pyplot as plt

def gaussian_kernel(x, xi, tau):

    return np.exp(-np.sum((x - xi) ** 2) / (2 * tau ** 2))

def locally_weighted_regression(x, X, y, tau):

    m = X.shape[0]

    weights = np.array([gaussian_kernel(x, X[i], tau) for i in range(m)])

    W = np.diag(weights)

    X_transpose_W = X.T @ W

    theta = np.linalg.inv(X_transpose_W @ X) @ X_transpose_W @ y

    return x @ theta

np.random.seed(42)

X = np.linspace(0, 2 * np.pi, 100)

y = np.sin(X) + 0.1 * np.random.randn(100)

X_bias = np.c_[np.ones(X.shape), X]

x_test = np.linspace(0, 2 * np.pi, 200)

x_test_bias = np.c_[np.ones(x_test.shape), x_test]

tau = 0.5

y_pred = np.array([locally_weighted_regression(xi, X_bias, y, tau) for xi in x_test_bias])
```

```
plt.figure(figsize=(10, 6))  
plt.scatter(X, y, color='red', label='Training Data', alpha=0.7)  
plt.plot(x_test, y_pred, color='blue', label=f'LWR Fit ( $\tau=\{\tau\}$ )', linewidth=2)  
plt.xlabel('X', fontsize=12)  
plt.ylabel('y', fontsize=12)  
plt.title('Locally Weighted Regression', fontsize=14)  
plt.legend(fontsize=10)  
plt.grid(alpha=0.3)  
plt.show()
```

**VIVA Questions with Answers after each experiment****1. What is Locally Weighted Regression (LWR)?**

Locally Weighted Regression is a non-parametric regression algorithm that fits a model locally around a query point using nearby data points.

**2. Why is LWR called a non-parametric algorithm?**

Because it does not assume a fixed global model or predefined parameters for the entire dataset.

**3. How does LWR differ from linear regression?**

Linear regression fits one global model, whereas LWR fits a local model for each query point.

**4. What role does the kernel function play in LWR?**

The kernel function assigns higher weights to nearby data points and lower weights to distant points.

**5. What is the bandwidth parameter in LWR?**

The bandwidth controls the size of the local neighborhood influencing the regression.

**6. Why is Euclidean distance commonly used in LWR?**

Euclidean distance measures the closeness between data points for weighting purposes.

**7. What happens if the bandwidth is too small?**

The model becomes sensitive to noise and may overfit the data.

**8. What happens if the bandwidth is too large?**

The model behaves like a global regression and may underfit the data.

**9. Why are graphs plotted in LWR experiments?**

Graphs help visualize how well the locally fitted curve matches the data points.

**10. What is a limitation of Locally Weighted Regression?**

LWR is computationally expensive because a new model is built for each prediction.

**Experiment 7.****Date:**

**AIM: Develop a program to demonstrate the working of Linear Regression and Polynomial Regression. Use Boston Housing Dataset for Linear Regression and Auto MPG Dataset (for vehicle fuel efficiency prediction) for Polynomial Regression.**

**Solution:**

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.preprocessing import PolynomialFeatures, StandardScaler

from sklearn.pipeline import make_pipeline

from sklearn.metrics import mean_squared_error, r2_score

def linear_regression_california():

    housing = fetch_california_housing(as_frame=True)

    X = housing.data[["AveRooms"]]

    y = housing.target

    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

    model = LinearRegression()

    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    plt.scatter(X_test, y_test, color="blue", label="Actual")
```

```
plt.plot(X_test, y_pred, color="red", label="Predicted")

plt.xlabel("Average number of rooms (AveRooms)")

plt.ylabel("Median value of homes ($100,000)")

plt.title("Linear Regression - California Housing Dataset")

plt.legend()

plt.show()

print("Linear Regression - California Housing Dataset")

print("Mean Squared Error:", mean_squared_error(y_test, y_pred))

print("R^2 Score:", r2_score(y_test, y_pred))

def polynomial_regression_auto_mpg():

    url = "https://archive.ics.uci.edu/ml/machine-learning-databases/auto-mpg/auto-mpg.data"

    column_names = ["mpg", "cylinders", "displacement", "horsepower", "weight", "acceleration",

                    "model_year",

                    "origin"]

    data = pd.read_csv(url, sep='\s+', names=column_names, na_values="?")

    data = data.dropna()

    X = data["displacement"].values.reshape(-1, 1)

    y = data["mpg"].values

    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

    poly_model = make_pipeline(PolynomialFeatures(degree=2), StandardScaler(),

                               LinearRegression())

    poly_model.fit(X_train, y_train)
```

```
y_pred = poly_model.predict(X_test)

plt.scatter(X_test, y_test, color="blue", label="Actual")

plt.scatter(X_test, y_pred, color="red", label="Predicted")

plt.xlabel("Displacement")

plt.ylabel("Miles per gallon (mpg)")

plt.title("Polynomial Regression - Auto MPG Dataset")

plt.legend()

plt.show

print("Polynomial Regression - Auto MPG Dataset")

print("Mean Squared Error:", mean_squared_error(y_test, y_pred))

print("R^2 Score:", r2_score(y_test, y_pred))

if __name__ == "__main__":

    print("Demonstrating Linear Regression and Polynomial Regression\n")

    linear_regression_california()

    polynomial_regression_auto_mpg()
```

**VIVA Questions with Answers after each experiment****1. What is Linear Regression?**

Linear Regression is a supervised learning algorithm that models a linear relationship between independent and dependent variables.

**2. Why is the Boston Housing dataset used for Linear Regression?**

Because it contains numerical features with approximately linear relationships to house prices.

**3. What is the target variable in the Boston Housing dataset?**

The median value of owner-occupied homes.

**4. What is Polynomial Regression?**

Polynomial Regression is an extension of linear regression that models non-linear relationships by adding polynomial terms.

**5. Why is the Auto MPG dataset suitable for Polynomial Regression?**

It shows non-linear relationships between vehicle features and fuel efficiency.

**6. How is Polynomial Regression implemented using Linear Regression?**

By transforming input features into polynomial features and then applying linear regression.

**7. What is the difference between Linear and Polynomial Regression?**

Linear regression models straight-line relationships, while polynomial regression models curved relationships.

**8. What happens if the polynomial degree is too high?**

The model may overfit the training data.

**9. Which evaluation metric is commonly used in regression problems?**

Mean Squared Error (MSE) is commonly used.

**10. What is one limitation of Polynomial Regression?**

It can become complex and overfit if the polynomial degree is not chosen properly.

**Experiment 8.****Date:**

**AIM: Develop a program to demonstrate the working of the decision tree algorithm. Use Breast Cancer Data set for building the decision tree and apply this knowledge to classify a new sample.**

**Solution:**

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score

from sklearn import tree


# Load the Breast Cancer dataset

data = load_breast_cancer()

X = data.data

y = data.target


# Split the data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)


# Train a Decision Tree Classifier

clf = DecisionTreeClassifier(random_state=42)
```

```
clf.fit(X_train, y_train)
```

```
# Make predictions on the test set
```

```
y_pred = clf.predict(X_test)
```

```
# Calculate and print the accuracy
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print(f"Model Accuracy: {accuracy * 100:.2f}%")
```

```
# Predict the class for a new sample (first sample in the test set)
```

```
new_sample = X_test[0].reshape(1, -1) # Reshape to match the input format
```

```
prediction = clf.predict(new_sample)
```

```
# Map the prediction to the class name
```

```
prediction_class = "Benign" if prediction == 1 else "Malignant"
```

```
print(f"Predicted Class for the new sample: {prediction_class}")
```

```
# Plot the decision tree
```

```
plt.figure(figsize=(12, 8))
```

```
tree.plot_tree(
```

```
    clf,
```

```
    filled=True,
```

```
    feature_names=data.feature_names.tolist(), # Convert NumPy array to list
```

```
class_names=data.target_names.tolist() # Convert NumPy array to list
)
plt.title("Decision Tree - Breast Cancer Dataset")
plt.show()
```

**VIVA Questions with Answers after each experiment****1. What is a Decision Tree?**

A Decision Tree is a supervised learning algorithm that uses a tree-like structure to make decisions based on feature values.

**2. Why is the Breast Cancer dataset used in this experiment?**

It is a standard dataset with well-defined features and class labels, suitable for classification tasks.

**3. What is the target variable in the Breast Cancer dataset?**

The target variable indicates whether the tumor is malignant or benign.

**4. How does a Decision Tree make predictions?**

It traverses from the root node to a leaf node based on feature conditions.

**5. What criterion is commonly used to split nodes in a Decision Tree?**

Gini Index or Entropy is commonly used as the splitting criterion.

**6. What is overfitting in Decision Trees?**

Overfitting occurs when the tree becomes too complex and fits noise in the training data.

**7. How can overfitting be reduced in Decision Trees?**

By pruning the tree or limiting the maximum depth.

**8. What is meant by classifying a new sample?**

It means predicting the class label of unseen data using the trained decision tree model.

**9. Which Python library is commonly used to implement Decision Trees?**

Scikit-learn is commonly used.

**10. What is one advantage of Decision Trees?**

Decision Trees are easy to understand and interpret.

**Experiment 9.****Date:**

**AIM: Develop a program to implement the Naive Bayesian classifier considering Olivetti Face Dataset for training. Compute the accuracy of the classifier, considering a few test data sets.**

**Solution:**

```
import numpy as np

from sklearn.datasets import fetch_olivetti_faces

from sklearn.model_selection import train_test_split, cross_val_score

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

import matplotlib.pyplot as plt


data = fetch_olivetti_faces(shuffle=True, random_state=42)

X = data.data

y = data.target


X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)


gnb = GaussianNB()

gnb.fit(X_train, y_train)

y_pred = gnb.predict(X_test)


accuracy = accuracy_score(y_test, y_pred)
```

```
print(f'Accuracy: {accuracy * 100:.2f}%')
```

```
print("\nClassification Report:")
```

```
print(classification_report(y_test, y_pred, zero_division=1))
```

```
print("\nConfusion Matrix:")
```

```
print(confusion_matrix(y_test, y_pred))
```

```
cross_val_accuracy = cross_val_score(gnb, X, y, cv=5, scoring='accuracy')
```

```
print(f'\nCross-validation accuracy: {cross_val_accuracy.mean() * 100:.2f}%')
```

```
fig, axes = plt.subplots(3, 5, figsize=(12, 8))
```

```
for ax, image, label, prediction in zip(axes.ravel(), X_test, y_test, y_pred):
```

```
ax.imshow(image.reshape(64, 64), cmap=plt.cm.gray)
```

```
ax.set_title(f'True: {label}, Pred: {prediction}')
```

```
ax.axis('off')
```

```
plt.show()
```

**VIVA Questions with Answers after each experiment****1. What is the Naive Bayes classifier?**

Naive Bayes is a probabilistic supervised learning algorithm based on Bayes' theorem with an assumption of feature independence.

**2. Why is the Naive Bayes classifier called “naive”?**

It assumes that all features are conditionally independent given the class label.

**3. What is the Olivetti Face Dataset?**

It is a dataset containing grayscale facial images of different individuals, used for face recognition tasks.

**4. How is the Olivetti Face Dataset used in this experiment?**

The dataset is used to train the Naive Bayes classifier to recognize different face classes.

**5. Why is Naive Bayes suitable for this dataset?**

It works well with high-dimensional data such as images and is computationally efficient.

**6. What type of Naive Bayes classifier is commonly used for image data?**

Gaussian Naive Bayes is commonly used for image data.

**7. What is meant by training and testing in classification?**

Training is learning from labeled data, while testing evaluates the model on unseen data.

**8. How is the accuracy of the classifier computed?**

Accuracy is computed as the ratio of correctly classified samples to the total number of test samples.

**9. What is one limitation of the Naive Bayes classifier?**

The independence assumption may not hold true for real-world data.

**10. What is one advantage of Naive Bayes?**

It is simple, fast, and works well with large datasets.

**Experiment 10.****Date:**

**AIM: Develop a program to implement k-means clustering using Wisconsin Breast Cancer data set and visualize the clustering result.**

**Solution:**

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.datasets import load_breast_cancer

from sklearn.cluster import KMeans

from sklearn.preprocessing import StandardScaler

from sklearn.decomposition import PCA

from sklearn.metrics import confusion_matrix, classification_report


data = load_breast_cancer()

X = data.data

y = data.target


scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

kmeans = KMeans(n_clusters=2, random_state=42)

y_kmeans = kmeans.fit_predict(X_scaled)
```

```
print("Confusion Matrix:")

print(confusion_matrix(y, y_kmeans))

print("\nClassification Report:")

print(classification_report(y, y_kmeans))

pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)


df = pd.DataFrame(X_pca, columns=['PC1', 'PC2'])

df['Cluster'] = y_kmeans

df['True Label'] = y


plt.figure(figsize=(8, 6))

sns.scatterplot(data=df, x='PC1', y='PC2', hue='Cluster', palette='Set1', s=100, edgecolor='black',
alpha=0.7)

plt.title('K-Means Clustering of Breast Cancer Dataset')

plt.xlabel('Principal Component 1')

plt.ylabel('Principal Component 2')

plt.legend(title="Cluster")

plt.show()


plt.figure(figsize=(8, 6))

sns.scatterplot(data=df, x='PC1', y='PC2', hue='True Label', palette='coolwarm', s=100,
edgecolor='black',
```

```
alpha=0.7)

plt.title('True Labels of Breast Cancer Dataset')

plt.xlabel('Principal Component 1')

plt.ylabel('Principal Component 2')

plt.legend(title="True Label")

plt.show()


plt.figure(figsize=(8, 6))

sns.scatterplot(data=df, x='PC1', y='PC2', hue='Cluster', palette='Set1', s=100, edgecolor='black',
alpha=0.7)

centers = pca.transform(kmeans.cluster_centers_)

plt.scatter(centers[:, 0], centers[:, 1], s=200, c='red', marker='X', label='Centroids')

plt.title('K-Means Clustering with Centroids')

plt.xlabel('Principal Component 1')

plt.ylabel('Principal Component 2')

plt.legend(title="Cluster")

plt.show()
```

**VIVA Questions with Answers after each experiment****1. What is k-means clustering?**

k-means is an unsupervised learning algorithm that groups data into k clusters based on feature similarity.

**2. Why is k-means called an unsupervised algorithm?**

Because it does not use labeled output data during training.

**3. What is the Wisconsin Breast Cancer dataset?**

It is a medical dataset containing numerical features of breast cancer tumors.

**4. What is the value of k used in this experiment and why?**

$k = 2$ , because the dataset has two natural groups: benign and malignant tumors.

**5. How does the k-means algorithm work?**

It initializes cluster centroids, assigns points to the nearest centroid, and updates centroids iteratively until convergence.

**6. Why is data normalization important before applying k-means?**

Normalization ensures that all features contribute equally to distance calculations.

**7. What distance measure is commonly used in k-means?**

Euclidean distance is commonly used.

**8. How are clustering results visualized in this experiment?**

Clustering results are visualized using scatter plots after dimensionality reduction.

**9. What is a limitation of the k-means algorithm?**

k-means is sensitive to the initial choice of centroids.

**10. How is clustering performance evaluated in this experiment?**

By comparing cluster labels with actual class labels or using metrics like silhouette score.

## **VIVA QUESTIONS**

- 1.What is data analysis.**
- 2.Why do we need to visualize data?**
- 3.Mention the differences between box plot and histograms.**
- 4.What is statistics?**
- 5.What is descriptive statistics.**
- 6.What is inferential statistics?**
- 7.Differentiate between Machine learning and AI.**
- 8.What are the different types of data.**
- 9.Illustrate the use of Histograms.**
- 10.Mentions the use of Box plots.**
- 11.What is IQR?**
- 12.How to handle missing values?**
- 13.What is EDA.**
- 14.Define outliers and hence how to we detect outliers.**
- 15.What is dimensionality reduction?**
- 16.Explain PCA.**
- 17.Differentiate between supervised and unsupervised learning.**
- 18.Differentiate between regression and classification.**
- 19.Give examples for bimary and multi classification.**
- 20.Give examples for univariate , multivariate and bivariate data analysis.**
- 21.Name the measures of central tendency.**

22. What is variance, bias and standard deviation.
23. What is heatmap?
24. Define correlation matrix.
25. Mention types of correlation
26. Explain the importance of pair plot.
27. Explain Find-s algorithm and its importance.
28. What do you mean by non-parametric algorithms.
29. Explain importance of KNN.
30. How do you compute the distance between data points.
31. Explain Decision tree.
32. What is information gain.
33. What is entropy.
34. What is Gini index.
35. What is Bayesian learning.
36. Differentiate between linear regression and polynomial regression.
37. Explain locally weighted regression.
38. Why do we use standard scaler?
39. Mention the applications of polynomial regression.
40. Mention the applications of clustering algorithms
41. What is elbow technique?
42. What is normalization?
43. List the applications of decision tree.
44. What is a dataset?

- 45. What is overfitting?
- 46. What is underfitting?
- 47. What is model accuracy?
- 48. What is supervised learning?
- 49. What is unsupervised learning?
- 50. What is training data and testing data?

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